



SIMT Engineering College
Dhaka University Affiliated Technology Unit
CSE 1201: Fundamentals of Programming
Incourse 2

Time: 1 hour 30mins

Marks: 30

(Write Any 6 Marks: $5 \times 6 = 30$)

Lo i' = 0,

1. Give an Example of Nested if-else using pseudocode and a flowchart.
2. Write a C code of Simple Calculator using Switch Statement.
3. Write a C program to check whether an alphabet is vowel or consonant using goto statement.
4. Write a C program to check whether a number is even or odd using ternary operator.
5. Write a C code to calculate the sum of first n natural numbers using for loop.
6. Write a program in C to display the pattern like right angle triangle using an asterisk
7. Give an Example and explain how do-while works.
8. How to Access Elements Out of its Bound in a single dimensional Array and what are the disadvantages of an Array.
9. Give an Example for Function without Argument and Return Value.
10. Difference between Call by Value and Call by Reference with Example of C code.

SAIC Institutes of Management and Technology

Affiliated Engineering Colleges, University of Dhaka

Department of Computer Science & Engineering

1st Year 2nd Semester In course -2 Examination, 2025

Course Code: CSE 1202

Time: 60 min

Course Title: Digital Logic Design

[Answer Any **Five** Questions]

Marks: 5*5=25

- 1/ Define Multiplexer and Implement the function $F(A, B, C, D) = \sum m(0, 1, 3, 4, 8, 9, 15)$ using an 8*1 Multiplexer. 5
- 2/ Simplify the following Boolean function $F(w, x, y, z) = \sum(0, 1, 2, 8, 10, 11, 14, 15)$ using the tabulation method. 5
- 3/ Draw the internal logic diagram of 32*8 ROM. 5
- 4/ A combinational circuit is defined by the functions $F_1(A, B, C) = \sum(3, 5, 6, 7)$ and $F_2(A, B, C) = \sum(0, 2, 4, 7)$. Implement the circuit with a PLA having three inputs, four product and two outputs. 5
5. Differentiate between Latch and flip-flop and also describe the SR flip-flop. 5
- 6/ Draw the block diagram with truth table of a 4*16 decoder using two 3*8 decoders. 5

00	00	01	00	01	11
01	00	01	00	01	11
10	00	01	00	01	11
11	00	01	00	01	11

$A \bar{B} \bar{C} D$
 $\bar{A} B \bar{C} D$

$\bar{B} \bar{C} (\bar{A} D + A D)$

SIMT Engineering College

Physics ii

CSE 1Y 2S

Answer all the following questions:

1. Distinguish between crystalline and non-crystalline solids.
2. Calculate the packing fraction for FCC unit cell.
3. Draw the crystal planes for $[1\ 3\ 1]$, $[1\ 0\ 1]$



Department of Computer Science & Engineering
SIMT Engineering College
M/3, Section-14, Mirpur, Dhaka-1216

Course Code: MATH 1204

Course Title: Method of Integration, DE & Series.

Exam - 01

Date: Nov 18, 2025

Time Allowed: 1h

Total Marks – 25

SL	Questions	Marks
Q1	An apple pie is taken out of an oven at a temperature of 200°F and placed on the counter in a room where the temperature is 70°F . The temperature of the pie is 150°F after 10 minutes. a. Find a relation that gives the temperature of the pie after t minutes. b. How long will it take for the pie to cool to 120°F ? c. Sketch the solution curve.	6
Q2	State and prove the necessary and sufficient condition for $M(x,y)dx + N(x,y)dy = 0$ to be exact.	5
Q3	Solve the following differential equations (any three) i. $xy - \frac{dy}{dx} = y^3 e^{-x^2}$ ii. $ydx + x(\ln x - \ln y - 1)dy = 0$ iii. $xdy - ydx = \sqrt{x^2 + y^2}dx$ iv. $\frac{dy}{dx} + \frac{y^2+y+1}{x^2+2x+1} = 0$	9
Q4	Define general and particular solution of a differential equations. Show that $y = (x^2 + c)e^{-x}$, where c is a constant, is the general solution of $\frac{dy}{dx} + y = 2xe^{-x}$. Find a particular solution for the condition $y(-1) = e + 3$.	5

SAIC Institute of Management and Technology
Affiliated Engineering College
University of Dhaka
Department of CSE

Course name: Developing English Language Skills
Course code: ENG 1205

Marks: 30
Time: 1 Hour

1. Suppose you have graduated from your department. Now prepare a cover letter and a CV for the following job circular:

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Company Name: NovoTech Engineering Ltd.
Position: Graduate Trainee Engineer
Location: Dhaka, Bangladesh
Vacancy: 05
Job Nature: Full-time
Salary: Negotiable
Application Deadline: November 20, 2025
Application with strong communication skills and excellent command over English will be given preferences. Please send your application to the following address:

*Mr. Mamunur Khan, Head of Human Recourse Department
NovoTech Engineering Ltd.
Banani, Dhaka.*

2. Write an email to the principal of your school requesting permission to arrange a cultural program

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3. Write an essay on "Causes and effects of stress related to future career uncertainty"

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